# ARTIFACT GATES AND BOUNDED VERIFIER FEEDBACK IN A TRACEABLE RESEARCH AGENT

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## ABSTRACT

A research agent can generate a plausible manuscript without leaving a complete, reviewable execution record. We implement an Artifact Provenance Gate inside JiuwenSwarm that checks a PDF, successful terminal Reporting, contained declared artifacts, aggregable model usage, and citation-key resolution when reliable LaTeX evidence exists. A specification-derived pilot rejects 20 injected blocking faults and accepts four valid controls; these fixtures test artifact integrity rather than scientific truth. We separately study whether one deterministic error diagnostic helps a model repair structured reasoning beyond generic reconsideration with the same retry opportunity. On 24 synthetic arithmetic, satisfiability, and shortest-path items, shared initial answers achieve 10/24 accuracy, generic retry 11/24, and diagnostic retry 12/24. The paired difference is one item, with diagnostic-only/generic-only discordance 1/0 and exact two-sided McNemar $p = 1.0$. The reasoning pilot is inconclusive. Its 52 actual calls consume 9,500 input and 888 output tokens. We disclose ten engineering executions on the same items, preserve failures, and distinguish the completed Agent manuscript from this independently compiled, operator-amended paper. Neither artifact validation nor repeated engineering runs establish a general reasoning improvement.

## 1 INTRODUCTION

Research agents connect planning, experiment execution, analysis, and writing. Their apparent endpoint is often a manuscript, but a readable document can coexist with missing runtime artifacts, unresolved citations, or unusable accounting. An operational completion flag therefore answers a narrower question than whether an experiment is reproducible or its conclusion is correct. We address this gap with a small deterministic completion contract implemented in the JiuwenSwarm framework, rather than a second language model judging another model's prose.

The contract asks whether observable evidence is present: a nonempty ordinary PDF, a successful terminal Reporting report, safe existing declared paths, a parsable usage ledger, and resolvable compiled citation keys. It blocks publication of completion when one of these conditions fails. It deliberately does not certify that a scientific claim follows from an experiment or that a cited paper supports a sentence. These distinctions prevent a successful system test from being presented as a scientific result.

For a real execution case, we examine externally assisted reasoning. Chain-of-thought prompting motivates attention to inference-time context (Wei et al., 2022), while iterative refinement raises a practical question: after an incorrect first answer, does a specific, automatically checkable error signal add value beyond simply asking the model to reconsider? Improvements over an initial answer alone confound information with an additional attempt. We therefore compare two selective retries that share the same initial answer and receive the same number of extra model calls. One adds a compact deterministic diagnostic; the other receives generic reconsideration text.

The resulting paired pilot is small and negative in its evidential conclusion: the diagnostic arm gains one item relative to the control, but a single discordant pair yields no significant paired evidence. The purpose is an inspectable measurement and system demonstration, not a claim of a new general reasoning algorithm. Our contributions are the framework completion gate, a traceable paired runner accepting legitimate alternative answers, and a bounded reporting integration that produces actual

ICLR artifacts while retaining failure records. Topic, sources, protocol, data, and executable code were prepared with operator assistance; the system did not autonomously discover or generate them.

## 2 Related Work

**Inference-time reasoning and revision.** Chain-of-thought prompting supplies intermediate reasoning examples as context (Wei et al., 2022). Our solver instead returns a strict structured answer; we neither reproduce chain-of-thought experiments nor compare explanation styles. Self-Refine iteratively uses model-generated feedback and revision (Madaan et al., 2023). Here the diagnostic is supplied by executable constraints, only one retry is permitted, and no learned feedback model is introduced. Huang et al. examine limitations of intrinsic reasoning self-correction without reliable external feedback (Huang et al., 2023). Their setting motivates separating unassisted reconsideration from external assistance; it does not predict a positive effect on our synthetic items.

**Tool-using and reflective agents.** ReAct combines reasoning and actions in interactive tasks (Yao et al., 2022); Reflexion uses verbal feedback in an agent framework (Shinn et al., 2023). These works contextualize agents that use environmental evidence. Our comparison concerns a compact diagnostic at a fixed answer-repair opportunity, not an unrestricted action loop or repeated verbal memory. The research pipeline's Reflection module analyzes experiment results; it is separate from the solver's diagnostic retry and should not be counted as another answer-repair arm.

**Evidence contracts.** The Artifact Provenance Gate evaluates local files and runtime state. Answer validation and artifact validation serve different roles: a SAT checker can accept any satisfying assignment, whereas the gate can only determine whether required reporting evidence exists and is interpretable. Citation-key resolution likewise proves linkage in a compiled document, not semantic citation support. We present a practical framework extension and controlled evaluation, without claiming that artifact gates, external feedback, or paired testing are previously unknown concepts. Related-work context was curated from primary-source metadata and abstracts; this paper does not imply a systematic review of all full texts.

## 3 Method

### 3.1 Framework completion contract

The implementation modifies JiuwenSwarm and uses the real openJiuwen Manager and execution/reporting modules. The default PAPER factory connects the existing local Provider to the artifact adapter. Direct completion, polled completion, and recovery of active tasks share a finalizer; persisted public status follows the task store rather than an earlier Provider snapshot. Existing failed tasks are preserved. A blocking decision retains the package and audit record instead of exposing a premature completed state.

The five gate rules are: APG001, a nonempty ordinary final PDF; APG002, parsable Manager state with successful terminal Reporting; APG003, existing declared artifacts contained within the allowed run root, with unsafe absolute or traversal paths rejected; APG004, parsable model-call records with unique identifiers and nonnegative integer input/output tokens that can be aggregated; and APG005, comparison of compiled citation and bibliography keys when reliable AUX exists. Without reliable AUX, APG005 is marked skipped, not passed. A blocker failure rejects completion. The gate uses no model calls. Audits at the run and package locations are each written atomically; this is not a cross-file transaction.

### 3.2 Paired reasoning protocol

The frozen, self-generated dataset has 36 items split into six development, six pilot, and 24 evaluation items. Evaluation includes eight items per family and 12 easy/12 hard labels. There is no training or weight update. Sequential arithmetic requires an exact integer. Satisfiable conjunctive normal form requires a full binary assignment; all satisfying assignments are accepted. Positive-weight graph tasks require a valid shortest path and a matching declared cost; tied optimal paths are accepted. Invalid JSON, missing fields, and invalid answer schemas remain failures rather than exclusions.

For evaluation item $i$, the initial answer is $a_i$. If it is correct, both final arms reuse it without an extra call. Otherwise each arm receives the problem, that same answer, and one reconsideration request. The diagnostic arm additionally receives the first applicable constraint error: an arithmetic mismatch category, an unsatisfied-clause index, an invalid graph/path condition, a path-cost mismatch, or a nonoptimal-path category. The diagnostic does not include the expected integer, a satisfying assignment, an optimal path, or its cost. The checker necessarily knows the verification oracle locally; that is different from forwarding an oracle answer to the solver. Error ordering and payloads are retained in the runner for inspection.

The control therefore matches extra *calls*, not tokens, wall time, or monetary cost. Both arms use one shared initial answer, and each initially incorrect item receives exactly one attempt per arm. Let $g_i, d_i \in \{0, 1\}$ denote final correctness. Our principal contrast is

$$\widehat{\Delta} = \frac{1}{24} \sum_{i=1}^{24} (d_i - g_i). \tag{1}$$

We report exact two-sided McNemar testing on $b = \sum_i d_i(1-g_i)$ and $c = \sum_i g_i(1-d_i)$: conditional on $b + c$, the null reference is a binomial distribution with probability $1/2$. Marginal Wilson intervals use $z = 1.96$ and $n = 24$; they describe accuracy uncertainty, not the paired difference. Initially correct answers are retained, so final accuracy cannot fall below initial accuracy by design.

### 3.3 Runtime and accounting

The solver uses the returned model identifier `deepseek-flash`, temperature zero, disabled thinking, a 2,048-output-token cap, non-streaming responses, and no automatic request retry. A returned identifier is not a frozen vendor-weight version. Each actual reply retains response identifier, content, finish reason, usage, cache-hit tokens, and client latency. Started/returned records describe one call and are not counted twice. The child runner appends measured usage under the public task identifier and separate Manager node reference; completed cache/resume records do not trigger new requests or duplicate entries. Missing usage remains a failure.

PreparedCode copies four frozen public files and runs an offline smoke check. A real execution subprocess then performs the study. Optional ICLR Reporting delegates to the upstream writer/verifier, exposes already-measured nested confidence bounds as scalar metrics for the unchanged checker, and materializes task-local styles and concise writing guidance. Once all seven sections exist, real compilation precedes bounded optional polishing. After a normal writer return, complete sections without a PDF permit one existing compile CLI invocation; absent materials, compiler errors, or timeout do not synthesize success. The installed SDK and gate rules remain unchanged.

## 4 Experiments

### 4.1 Reasoning results

We use the first scientific execution of the tenth engineering run as the primary result because it accompanies the first completed public workflow with this frozen reporting implementation. We did not select it by accuracy. Earlier executions reused the same evaluation set during engineering diagnosis; the tenth run is not a fresh, blinded test or ten independent samples. An independent review recomputes answer validity, paired counts, intervals, usage, and parent/child record correspondence from retained replies. All 24 items remain in each denominator; all 52 returned calls have valid structured answers and finish reason `stop`.

Diagnostic-minus-generic accuracy is $1/24 = 0.0417$. Both arms are correct on 11 items and incorrect on 12; exactly one item is diagnostic-only correct and none generic-only correct. Thus $b = 1, c = 0$ and exact two-sided $p = 1.0$. A single discordance cannot provide conventional significance under this test. This is inconclusive evidence, not proof of no effect or equivalence. Relative to the shared initial answer, generic repairs one of 14 failures (graph-11), and diagnostic repairs two (graph-07 and graph-11). Repair counts 1/2 and discordance counts 1/0 describe different comparisons.

Both repairs occur on easy graph items; there is no observed repair in arithmetic, SAT, or hard items (Table 2). Of the 14 generic retries, ten reproduce the initial raw text and 11 reproduce its parsed

| Arm | Correct/24 | Accuracy | Wilson 95% interval |
|---|---|---|---|
| Shared initial | 10 | 0.4167 | [0.2447, 0.6117] |
| Generic retry | 11 | 0.4583 | [0.2789, 0.6493] |
| Diagnostic retry | 12 | 0.5000 | [0.3143, 0.6857] |

Table 1: Marginal accuracy on the same 24 items. Initial correct answers are retained in both retry arms. Intervals do not test the paired contrast.

| Group | $n$ | Initial | Generic | Diagnostic |
|---|---|---|---|---|
| Arithmetic | 8 | 3 | 3 | 3 |
| SAT | 8 | 4 | 4 | 4 |
| Graph | 8 | 3 | 4 | 5 |
| Easy | 12 | 8 | 9 | 10 |
| Hard | 12 | 2 | 2 | 2 |

Table 2: Correct counts, independently recomputed by family and difficulty. These small post-execution descriptions are not separate confirmatory tests.

JSON. One shared repair and three text changes that remain incorrect show why text change cannot stand in for reasoning improvement. The initial first-error categories are five arithmetic mismatches, four unsatisfied clauses, three graph cost mismatches, and two nonoptimal paths. The final diagnostic arm retains 12 errors, not all 14 initial errors.

### 4.2 Measured solver resources

The 52 actual calls decompose as $24 + 14 + 14$, below the maximum $24 + 2 \times 24 = 72$. Input/output tokens total 9,500/888, summed client time is 40.8030 seconds, and experiment elapsed time is 41.3663 seconds. All 52 replies match the parent ledger's corresponding input, output, and cache counts. Cache-hit tokens total 1,536 and are a subset of input, not an additional quantity.

### 4.3 Separate artifact-gate pilot

A specification-derived fault-injection set contains four valid controls and 20 annotated blockers, four assigned to each gate rule. The historical full gate yields 20 true positives, four true negatives, and no false positives/negatives on this set. A defined terminal-trusting policy accepts every injected fault; a PDF-only baseline detects four of 20. Thirty evaluations per fixture yield 720 recorded timed evaluations, with median 11.58445 ms and p95 23.923825 ms. These are repeated timings of 24 fixtures, not 720 independent defects; retained aggregates lack the 720 raw samples and bind an earlier gate, not current-runtime speed. For the 20 ledgers valid enough to aggregate, the before/after unique-call token delta is zero. Leave-one-rule decisions posthoc filter the full audit rather than rerun a reduced gate. APG002/APG003 fixtures have coupled failure behavior, so this cannot establish that APG002 is redundant. This controlled coverage does not establish resistance to arbitrary corruption, scientific correctness, or reasoning gains.

## 5 Discussion

### 5.1 Interpretation and limitations

The paired control isolates an additional diagnostic from an additional answer attempt at a matched call count. The observed advantage is one easy graph item. A missing generic-only success in this run is an observation, not a guarantee that diagnostic context cannot harm a retry. We make no population-level improvement claim. Non-significance does not establish a null effect, and overlapping marginal intervals are not a substitute for paired testing. No formal prospective power analysis was performed; the small number of observed discordances limits inference without proving that significance was impossible before execution.

| Actual requests | Calls | Input | Output | Client seconds |
|---|---|---|---|---|
| Shared initial | 24 | 3,644 | 389 | 18.7326 |
| Generic retry | 14 | 2,804 | 252 | 11.6903 |
| Diagnostic retry | 14 | 3,052 | 247 | 10.3802 |

Table 3: Solver requests only. Retry arms have equal call opportunities, but diagnostic uses 248 more input and five fewer output tokens. Timing is descriptive, not a controlled speed comparison.

| Run | Initial/Generic/Diagnostic | Calls | Input | Output | Public state |
|---|---|---|---|---|---|
| 1 | 10/11/12 | 52 | 9,504 | 892 | Failed |
| 2 | 9/11/12 | 54 | 9,865 | 919 | Failed |
| 3 | 10/11/12 | 52 | 9,504 | 888 | Failed |
| 4 | 10/11/12 | 52 | 9,504 | 891 | Failed |
| 5 | 10/11/12 | 52 | 9,504 | 890 | Failed |
| 6 | 10/11/12 | 52 | 9,496 | 879 | Failed |
| 7 | 9/10/12 | 54 | 9,739 | 904 | Failed |
| 8 | 9/11/12 | 54 | 9,865 | 920 | Failed |
| 9 | 10/11/12 | 52 | 9,504 | 892 | Failed |
| 10 | 10/11/12 | 52 | 9,500 | 888 | Completed |

Table 4: Engineering repetitions on the same 24 evaluation items. Correct counts and solver-only resources remain separate from full-workflow status. Full failure records are retained.

The study is restricted to one model identifier, one configuration, one retry, and three synthetic verification families. The graph checker can accept alternate shortest paths and the SAT checker alternate satisfying assignments, but this does not make the dataset representative of natural-language or real research problems. Difficulty labels are generator labels, not calibrated psychometric measurements. Repeated engineering exposure and vendor variability further constrain interpretation; temperature zero does not ensure identical replies or requests across executions. The diagnostic provides external assistance, not intrinsic self-correction or weight learning (Huang et al., 2023; Madaan et al., 2023).

### 5.2 Engineering disclosure and manuscript provenance

Ten paid engineering executions of the same 24 items are retained in chronological order (Table 4). Their 526 solver calls, 95,985 input and 8,963 output tokens are cumulative resources, not 240 independent observations or a pooled accuracy estimate. Runs 1–9 publicly failed, including cases where Reporting produced a PDF but accounting or Manager termination blocked completion. Run 9 performed Reflection after both Reporting failures; an incorrect statement in that raw Reflection about all 14 failures remaining unrepaired is preserved as an error, not copied into our analysis.

The tenth full workflow took 924.82 seconds, with successful Reporting lasting 812.046 seconds. Its original production audit passes all five rules without skips. Its 145 known-usage ledger entries aggregate 4,586,970 input and 67,067 output tokens; 4,421,760 cache-hit tokens are included in input. These full-workflow quantities differ from solver resources. The inner writer nevertheless reached its 64-step limit; the actual compiler and upstream output verification produced a valid artifact and successful outer Reporting. This is artifact completion with a disclosed inner-session limitation, not clean autonomous writer termination.

The historical run10 framework had 384 distinct offline regression checks (366 RSI, 11 solver, two legacy repair, five reconciliation), normal exits, and an independently applicable 20-file patch. These checks do not validate semantic claims or the later implementation. This manuscript updates the operator-amended copy of the completed Agent manuscript: wide generated tables were replaced with compact measured tables, duplicate bibliography authors corrected, and overstatements removed. Source/PDF hashes and each changed file are retained; original outputs are unchanged. Fresh held-out items, more model/configuration replications, and a length-matched retry control were identified as necessary before drawing a stronger conclusion. Appendix A separately identifies later evidence without changing this pilot's results.

## 6 CONCLUSION

We implement a deterministic artifact-completion gate within JiuwenSwarm and document a historical real, traceable research workflow with an independently reviewed ICLR manuscript. Its controlled artifact pilot identifies all 20 specified blocking faults but does not certify scientific truth. A separate paired reasoning pilot compares one external diagnostic retry against one generic retry on the same 24 synthetic items. Diagnostic accuracy is 12/24 versus 11/24 for generic retry, with discordance 1/0 and exact $p = 1.0$: the evidence is inconclusive. Complete records, accepted alternative answers, unequal token costs, failed engineering runs, and operator amendments are disclosed. Broader reasoning enhancement remains an empirical question requiring larger, fresh evaluations.

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# A CURRENT EVIDENCE AND HISTORICAL BOUNDARIES

This maintenance revision preserves the original 24-item scientific result and title. It does not retrospectively assign later source validation or scientific responses to the historical workflow. Full records are in the local evidence bundles; this appendix states the version and inference boundaries.

**Framework and artifact scope.** The current implementation has seven framework patches over the fixed upstream commit, 24 modified paths, and 4924 files. Independent reconstruction succeeds in 14 check/apply commands; 4602 text files are compared with CRLF-to-LF normalization only, and 322 binary files raw. The current 432-case RSI suite yields 430 PASS and two Windows symbolic-link SKIPPED cases. The historical 384 checks in the main text, and later 380 checks, are not accumulated into that result. The 24-fixture pilot binds an earlier gate version. Transport dropped two empty fixture directories; restoring the original entries yields 24/24 canonical identities with all 143 file payloads unchanged. This is not a new gate or timing run; archives must retain the empty directories. A separate 27-case scope probe uses gate bytes identical to the current gate: two controls pass, 18 in-scope defects block, and seven out-of-scope cases pass. Missing AUX means APG005 SKIPPED. Other boundary cases expose scientific truth, ledger completeness, and stale AUX limitations; PASS is not semantic certification. The 27 cases are not combined with the 24 fixtures as independent production faults.

**Actual workflows.** Historical run13 completed in 510.406 controller seconds, with Design, PreparedCode, Execution, and Reporting, no independent Reflection, and five production APG passes. Run16, on a six-patch source, failed in 423.6493584 seconds: 80 successful model responses and three local budget refusals. Outer Reporting reported success despite an inner writer error; subsequent Manager decisions were blocked. Its compiled PDF is retained, but APG was not run; sealed verification has 13 PASS and four FAIL checks. The later Reporting-status fix propagates explicit session errors and passes six installed-SDK cases offline; current seven-patch source has no new successful paid full workflow. APG checks terminal Reporting evidence after successful Manager outcome; it does not independently certify all Manager semantics. Offline replay, compilation, and source tests are not new Agent completion.

**Separate scientific evidence.** The later studies keep their own planned populations and controls: 592/600 complete pairs under a decoded-length control, 271/375 under actual input-count matching on reused historical initial errors, and 102/240 on new tasks. Their diagnostic/control discordances are respectively 32/2, 30/4, and 3/1. Full-population possible-outcome bounds are [0.0367,0.0633], [-0.2080,0.3467], and [-0.5667,0.5833]; these are not confidence intervals. The oldest contrast retains an input-token confound; the reused cohort is selected and incompletely covered; the new cohort has exact $p = 0.625$ with 138 missing pairs. Input-count equality does not imply semantic or compute equality. Coverage and residual-covariate descriptions are posthoc, not causal adjustment. These studies are neither pooled with each other nor added to the original 24-item pilot. They do not change its inconclusive conclusion or establish general reasoning improvement.